# UNDERSTANDING CHILDREN'S GOSSIP FROM DAILY TALK USING AN LLM-ASSISTED FRAMEWORK

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## Background

- **Gossip** refers to evaluative information about **absent third parties** and serves important social functions in communication and reputation sharing (Dunbar, 2004 ; Xu et al., 2011) .
- **Young children are capable of** gossiping (Engelmann et al., 2016 ), but empirical evidence about **how gossip appears in everyday interactions** remains limited.
- Traditional qualitative coding of conversational data is **time-consuming and labor-intensive**, limiting large-scale analysis.
- **Large Language Models (LLMs)** offer a potential tool to assist in identifying and coding conversational transcript (Long et al., 2024) .

## Research Question

1. Can a **Large Language Model (LLM)-assisted framework** reliably identify children's gossip in naturalistic conversations?
2. What are the **frequency, targets, and emotional valence** of young children's gossip in everyday interactions?

## Methods

**Study 1: Framework Development and Validation**

- Based on Qwen-Plus, developed an LLM-assisted coding framework for identifying children's gossip.
- Data source: Mandarin Zhou Dinner-talk Corpus from the Child Language Data Exchange System, focusing on children's gossip during family dinner conversations (Li & Zhou, 2015 ).
- Participants: 5 to 6-year-olds (N=72)
- Compared LLM-generated coding with human-coded data to evaluate reliability and performance.

**Study 2: Analysis of Naturalistic Dialogue Data**

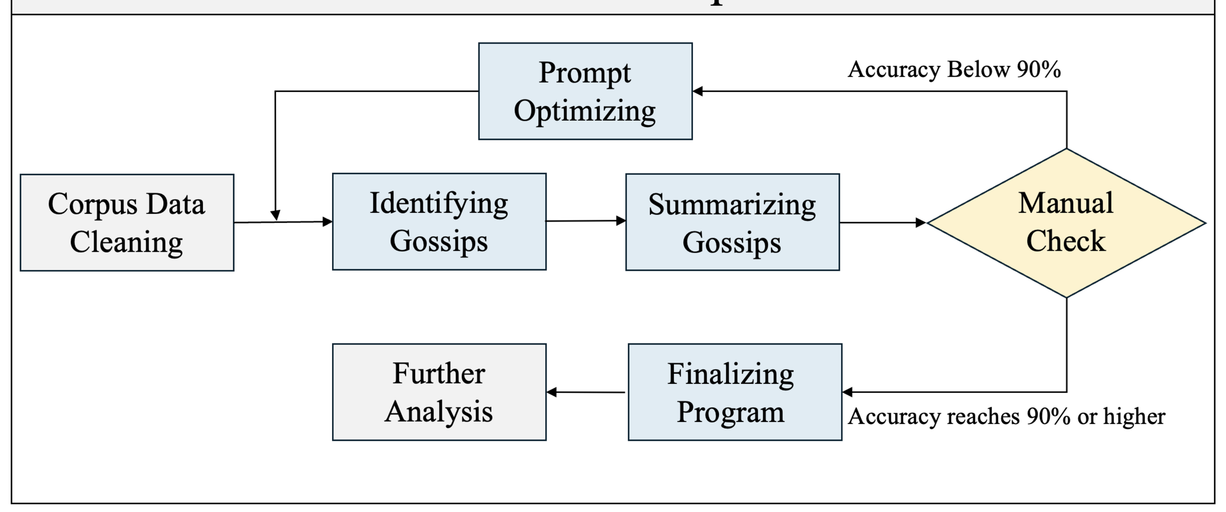
- Collected **daily conversational recordings** from children using portable audio-recording devices.
- Participants: **5-year-olds (*N* = 6)** and **6-year-olds (*N* = 3)**
- Recorded **children's everyday interactions with parents and peers**.

**Analytical Focus**

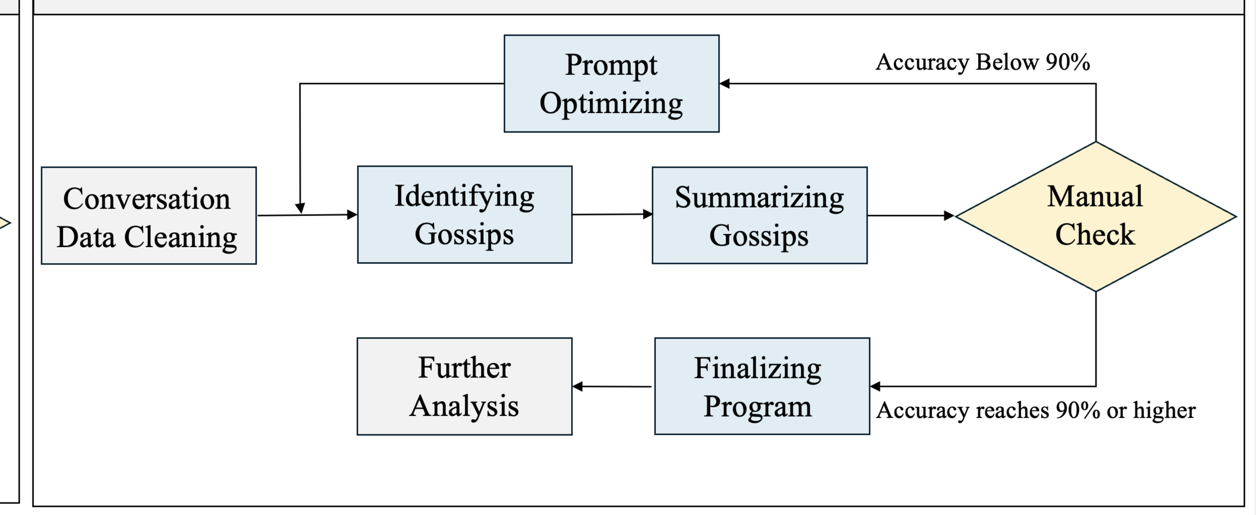
- Analysis of gossip coding quality of LLM-assisted framework: **Accuracy; Precision; Recall; F1 Score**
- Analysis of gossip characteristics: **Context; Frequency; Targets of gossip; Valence**

## The Procedure of LLM Program Development

Study 1: Developing an LLM Program Based on Corpus

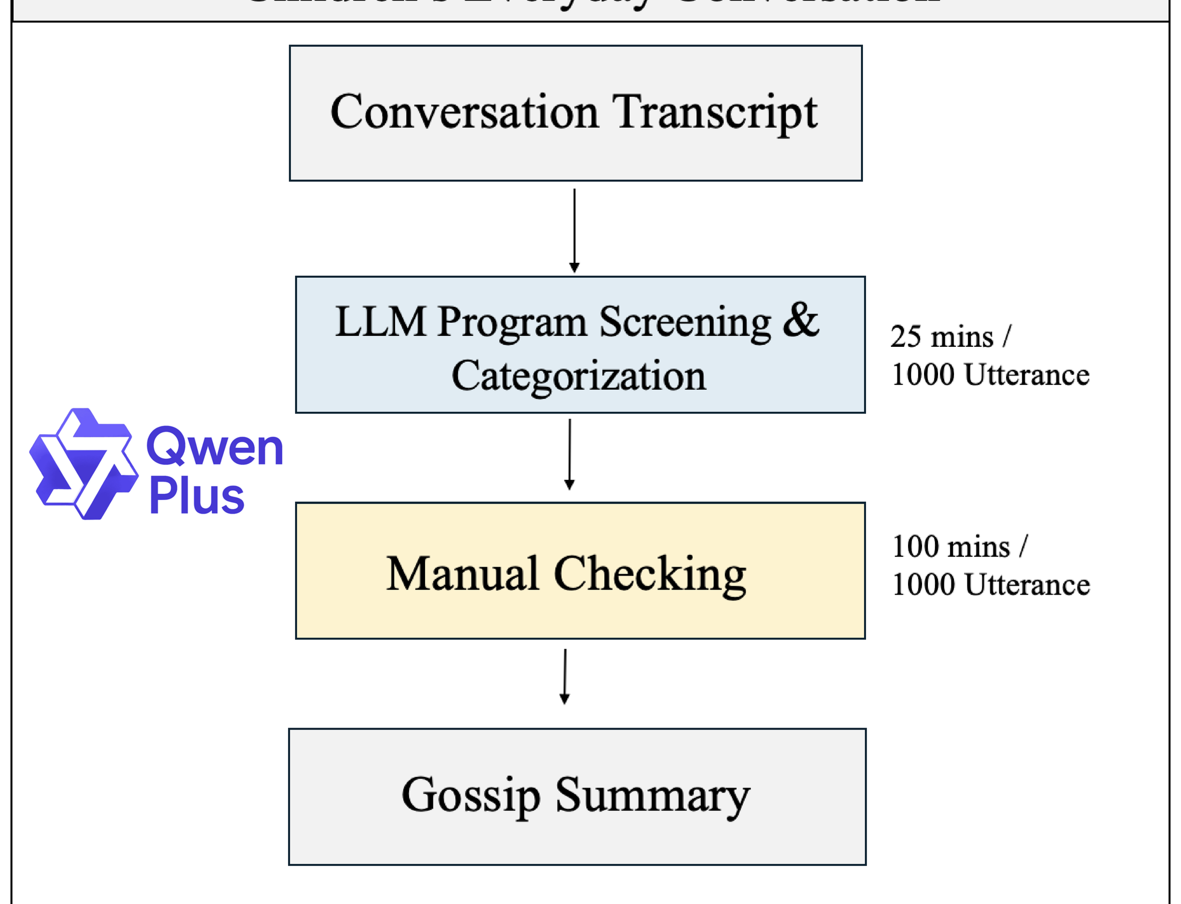



Study 2: Analysis of Gossip in Children's Everyday Conversation

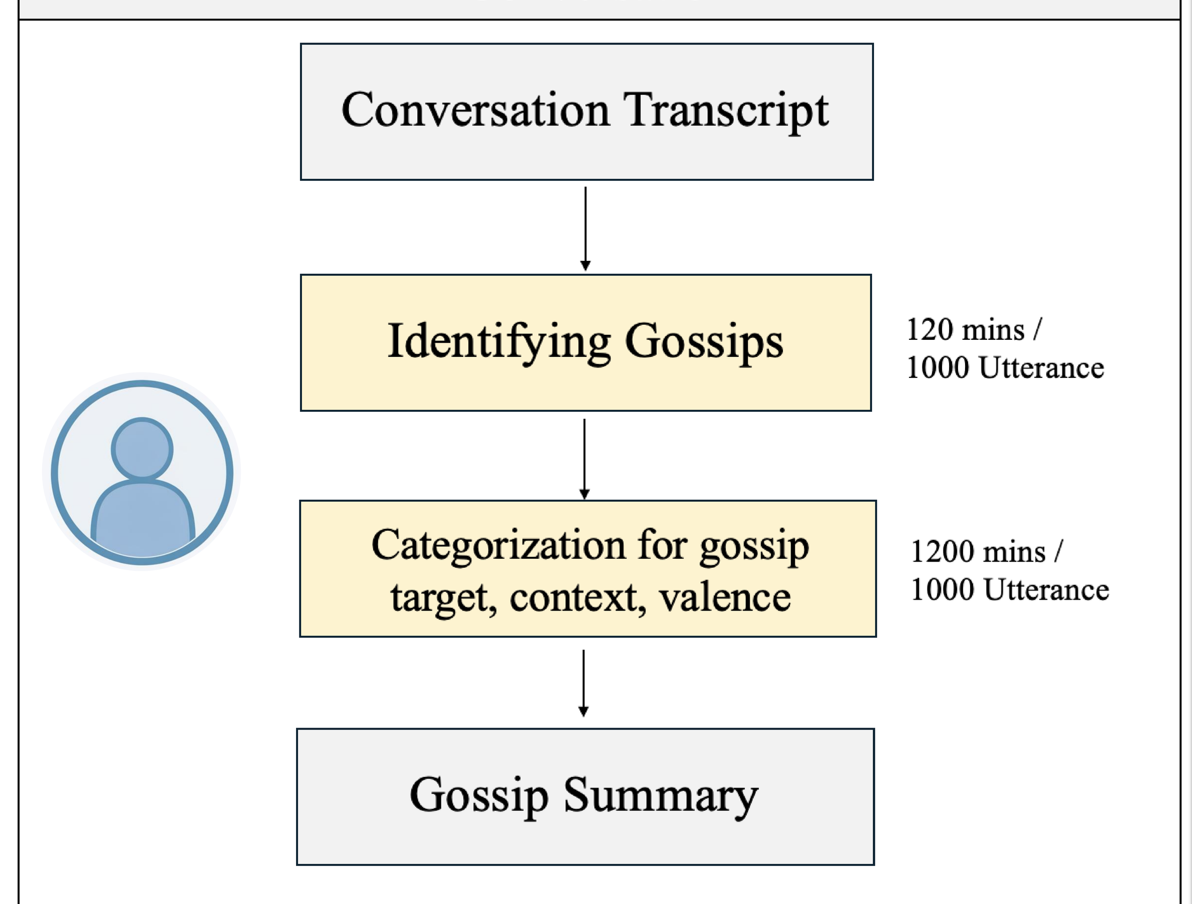



## The LLM-Assisted Framework V.S. Manual Coding in Children's Gossip Identification and Classification

The LLM-Assisted Framework Coding Corpus / Children's Everyday Conversation




Manual Coding on Corpus / Children's Everyday Conversation




**10.56 times FASTER**

**How Accurate Can Coders Code Children's Gossip? （Study 1）**

| Coder | Accuracy | Precision | Recall | F1 score |
|---|---|---|---|---|
| Human Coder 1 | 98.11% | 93.35% | 82.38% | 87.52% |
| Human Coder 2 | 98.07% | 90.92% | 76.95% | 83.35% |
| LLM (qwen-plus) | 93.78% | 70.35% | 71.30% | 70.82% |
| LLM-Assisted Framework | 96.84% | 98.76% | 71.14% | 82.70% |

**How Consistent Can Coders Code Children's Gossip? (Cohen's Kappa)**

| Cohen's Kappa | Study 1 | Study 2 |
|---|---|---|
| Coder 1 & Coder 2 | 0.76 | 0.90 |
| LLM Program & Human | 0.57 | 0.35 |
| LLM-Assisted Framework & Human | 0.82 | 0.86 |

## Results

**Children's Gossip Target and Valence in Everyday Conversation**

| Valence | Family | Peer/ Friends | School/ Teacher | Social Figure | Food | Others | Sum |
|---|---|---|---|---|---|---|---|
| Positive | 1 | 8 | 2 | 8 | 5 | 5 | 29 |
| Negative | 5 | 30 | 2 | 0 | 1 | 5 | 42 |
| Neutral | 21 | 38 | 23 | 0 | 13 | 31 | 121 |
| Sum | 27 | 76 | 27 | 8 | 19 | 41 | 192 |

**The Development of Children's Gossip Behavior**

| Age Group | Utterance | Gossips | Frequency of Gossip |
|---|---|---|---|
| 5-year-old | 1029 | 122 | 11.86% |
| 6-year-old | 549 | 73 | 13.30% |

**Children's Gossip Frequency in Difference Context**

| Context | Utterance | Gossip | Frequency of Gossip |
|---|---|---|---|
| Child-Parent | 484 | 82 | 16.94% |
| Child-Teacher | 215 | 21 | 9.77% |
| Child-Peer | 879 | 92 | 10.47% |

## Conclusion & Discussion

**Can an LLM-Assisted Framework Help Us Understand Children's Gossip?**

- The LLM-assisted framework showed strong accuracy and greatly improved coding efficiency, demonstrating its value for large-scale analysis of conversational data.

**The Characteristics of Children's Gossip**

- Children's **gossip frequency increased from age 5 to 6**, suggesting **early developmental changes** in social information sharing.
- Gossip occurred **more often in Child-Parent conversations** than in Child-Peer/Child-Teacher conversation, highlighting the role of family contexts in early gossip behavior.
- **Most gossip was neutral**, but negative gossip appeared more often about peers, reflecting emerging peer evaluation in childhood.

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